



FEEDBACK

YOUR SUGGESTIONS

ARTIFICIAL INTELLIGENCE

This is regarding the information on artificial intelligence published under 'Feedback' page of August issue. Artificial intelligence (AI) and machine learning (ML) often crop up simultaneously in topics like big data and analytics. Both of these are often used interchangeably, yet they differ from each other. ML is basically a subset of AI. On the one hand, AI is the intelligence exhibited by machines by which these mimic cognitive functions that humans use while interacting with other humans. ML, on the other hand, is the ability of machines to learn for themselves from the data provided to them.

Sagar Raj

Through e-mail

EFY. Thanks for the feedback and your views on AI and ML.

BENCH POWER SUPPLY

In 'Bench Power Supply Using a Computer's Power Supply' DIY article published in August issue, it is important to add a small load to keep the SMPS turned on always. A 10-ohm, 10W resistor can be connected across 5V rail and ground for the purpose.

Shashank T.S

Through e-mail

□ The 'Bench Power Supply Using a Computer's Power Supply' is a very nice project. I am using such a power supply for my experiments at home. Since it can give a large current, you need to protect the circuit by using automobile DC fuses (12V, 1-4A, etc).

Mahesh Gokhale

Through e-mail

The author Jaywardhan Pandit replies:

Thanks for your comments. The switch-mode power supply (SMPS) I am using is working fine without a load resistor connected to it. Load resistor is not required in all the SMPS. However, it can be used if the user faces problem.

Automobile fuse can be used if it is faster than the normal fuse.

VIBRATION SENSOR

I liked the 'Vibration Sensor' DIY article published in February issue. How can I extend the project to some other applications?

Sneha Reddy

Through e-mail

EFY. You can use this project for smart home systems, automotive devices, communication devices, electronic scales, instruments/toys, sports equipment, etc. The circuit design, though, depends on the intended application. For example, you can replace the buzzer and its driver section with an audio amplifier and speaker to use it as an anti-theft system for your vehicle.

3W/6W AUDIO AMPLIFIER

In '3W/6W Audio Amplifier Using TDA2003' circuit published in June issue, can we use TDA2030 IC? What are the changes required if we replace TDA2003 with TDA2030?

Sayantan Maiti

Through e-mail

EFY. There are slight differences between the two ICs. Apart from their output wattage differences, gain settings are also different. You need to refer to TDA2030's datasheet to

configure its pins for optimal component values for maximum outputs.

FLASH TRIGGER

In the PCB layout of 'Optical Slave Flash Trigger' DIY project published in August issue, pin 2 of IC1 (MOC3021) is not connected to the collector of T2 (BC547). So it will not work as described.

Tech Crafts

Through e-mail

EFY. Thanks for pointing out the mistake!

3W AUDIO AMPLIFIER

In '3W Audio Amplifier Using TDA7056' DIY project of July issue, a good-quality power transformer along with a full-bridge rectifier and a 2200µF capacitor will give you clean and stable 12 volts without a 7812 voltage regulator. To prevent overheating of the transformer, it should be rated 5 amps or more, so that TDA7056 consumes 50 per cent or so of the transformer's full capacity.

Tom Show

Through e-mail

EFY. The transformers that we purchased from the local market were not so reliable and exceeded their rated voltages. So to protect TDA7056 from high voltages, 7812 voltage regulator was used in this circuit. Also, use of a 7812 IC is much more cost-effective than a good-quality transformer. Finding a good-quality transformer in the local market is not easy. However, if you manage to get one with proper ratings, you may use a full-bridge rectifier with a zener diode across it.